

Arbitraging the Information Diffusion Lag: A Market-Neutral Statistical Arbitrage Strategy using Continuous Prediction Markets

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Abstract

This paper explores a novel statistical arbitrage strategy that capitalizes on the information diffusion lag between continuous prediction markets (e.g., Kalshi) and discrete traditional equity markets. By extracting real-time interest rate probability shifts ($\Delta\tau_t$) and constructing a market-neutral portfolio ($\beta_M = 0$) optimized for interest rate sensitivity (β_τ), we isolate the macroeconomic signal from broad market noise. We present the theoretical framework, mathematical portfolio construction, target universe selection, system architecture, and risk mitigation strategies for high-latency environments.

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1 Introduction

1.1 Continuous vs. Discrete Information Markets

Traditional macroeconomic trading heavily relies on discrete data releases and static consensus estimates (e.g., Bloomberg consensus). In contrast, modern prediction markets function as continuous probability density functions (PDFs). They aggregate dispersed information in real-time, frequently preempting formal announcements and traditional analyst revisions [1]. This continuous stream of belief updates provides a high-fidelity, leading macroeconomic indicator.

1.2 The Information Diffusion Lag Hypothesis

When a macroeconomic shock occurs, high-frequency trading (HFT) algorithms and Exchange Traded Fund (ETF) arbitrageurs instantly adjust broad market indices (e.g., SPY or QQQ). Da and Shive (2018) demonstrate that this ETF-induced activity creates excessive, non-fundamental comovement among constituent stocks [2]. In other words, index constituents move together purely based on their inclusion in the basket, temporarily overriding their specific fundamental sensitivities to the shock.

To illustrate the empirical findings of Ai et al. (2021)—which prove that equities possess highly heterogeneous sensitivities to monetary policy [5]—consider a prediction market signaling an unexpected interest rate hike. Index arbitrageurs might push the entire QQQ ETF down, causing both Alphabet (GOOG) and Tesla (TSLA) to drop by an identical -1.5% initially. However, fundamental corporate finance dictates that a cash-rich company like GOOG possesses a near-zero or positive interest rate beta, while a heavily leveraged, capital-intensive company like TSLA possesses a deeply negative beta [5].

Boguth et al. (2023) document that such aggregate price pressures and index-level noise eventually reverse as information is fully digested [6]. Building on these proven mechanisms, we hypothesize that a measurable "Information Diffusion Lag" exists: it takes minutes to days for human analysts and fundamental models to correct this blind indexing, leading to a delayed, exploitable cross-sectional repricing between the constituents.

1.3 Research Objectives

As established by Diercks et al. (2026), prediction markets do not explicitly signal when to execute equity rebalancing; rather, they act as a leading macroeconomic oracle [3]. Once this macroeconomic shift is visible, microstructure mechanisms ensure that broad indices are repriced almost instantaneously by HFTs [2]. This rapid aggregate adjustment effectively eliminates the opportunity to competitively trade the index itself.

Therefore, this research seeks to exploit the remaining inefficiency: the information diffusion gap left behind in individual constituent stocks [5]. Based on the empirical validation that prediction markets display the highest signal-to-noise ratio in forecasting inflation [3], our empirical testing hierarchy will prioritize Consumer Price Index (CPI) shocks, followed by Federal Reserve rate decisions (FED), and finally Non-Farm Payrolls (NFP). We aim to demonstrate that these early signals can trigger profitable, medium-frequency long/short positions across individual equities sorted by their specific macroeconomic betas.

2 Theoretical Foundation and Literature Review

2.1 Prediction Markets as Continuous Probability Density Functions

Prediction markets have been empirically validated as superior aggregators of information compared to polls or expert surveys. Recent studies on platforms like Kalshi demonstrate their efficacy in forecasting macroeconomic variables, such as CPI and Federal Reserve target rates, offering a real-time, distributional view of market expectations [3].

2.2 Microstructure Frictions: Why Direct PM Trading Fails

While the informational value of prediction markets is high, executing algorithmic trading directly within these platforms is highly constrained. Thin order books, wide bid-ask spreads, and the "favourite-longshot bias" [4] introduce severe liquidity frictions and slippage. Therefore, this paper utilizes prediction markets strictly as an information oracle, routing the execution to the highly liquid traditional equity markets.

2.3 Information Asymmetry in Equity Cross-Sections

The propagation of ETF-induced liquidity shocks creates excessive comovement among stocks [2]. However, as information diffuses, markets eventually price in firm-specific macro sensitivities. Studies show that portfolios sorted on monetary policy announcement premiums yield significant returns [5], proving that the cross-section of equities contains delayed, exploitable responses to macro shocks [6].

3 Mathematical Framework and Portfolio Construction

3.1 Intuition: The Logic Behind the Math

Before diving into the formal equations, let us explain the core logic in plain terms. When prediction markets signal a rate hike, HFT algorithms instantly crash the entire stock index. If our algorithm simply tries to "short" the market, we will lose, because the HFTs will always execute their trades milliseconds before us. We cannot beat them at guessing the market's immediate direction.

To survive and profit, we must step outside the HFTs' game. Instead of betting on the whole market crashing, we bet on the *relative difference* between two stocks after the dust settles. We place a "Long" (buy) bet on a cash-rich company like GOOG, and a "Short" (sell) bet on a debt-heavy company like TSLA.

This is where the mathematics below come into play. Their primary job is to calculate the exact dollar amount we need to put into GOOG and TSLA so that the HFT-driven market crash does not affect our wallet at all. If the HFTs push the entire market down by 2%, our GOOG long position loses money, but our TSLA short position makes the exact same amount of money. They perfectly cancel each other out. This is what "market neutrality" or "zero market exposure" means in practice: we make ourselves completely immune to the immediate, violent wave caused by the HFTs.

Our profit comes exclusively from what happens next (the Information Diffusion Lag). Over the following hours, human investors and fundamental algorithms realize that TSLA is fundamentally hurt by the rate hike, while GOOG is perfectly fine. Consequently, TSLA drops further while GOOG recovers. Because our mathematical constraints immunized us against the initial HFT crash, 100% of our profit is generated by this delayed, fundamental divergence between the two companies. The equations below simply formalize how to build this exact shield and profit engine.

3.2 The Macro-Factor Pricing Model

We assume that ETF arbitrage propagates macroeconomic shocks uniformly across index constituents in the immediate short-term (T_0). We model the return of an individual stock i at time t as a multi-factor process:

$$R_{i,t} = \alpha_i + \beta_{i,M} R_{M,t} + \beta_{i,\tau} \Delta\tau_t + \epsilon_{i,t} \quad (1)$$

Where $R_{M,t}$ is the broad market return, $\Delta\tau_t$ is the real-time shock to macroeconomic expectations, $\beta_{i,M}$ is the traditional market beta, and $\beta_{i,\tau}$ is the stock-specific sensitivity to interest rate expectations.

3.3 Isolating Interest Rate Beta (β_τ)

We estimate $\hat{\beta}_{i,\tau}$ using a rolling multivariate regression on historical prediction market data over a lookback window L :

$$\hat{\beta}_{i,\tau} = \frac{Cov(R_i, \Delta\tau)}{Var(\Delta\tau)} \Big|_{t-L}^t \quad (2)$$

3.4 The Continuous Signal Generator ($\Delta\tau_t$)

Let $P_{PM}(t)$ be the implied probability of a macroeconomic event on the prediction market. The signal is generated when the probability shift exceeds a predefined noise threshold (γ):

$$\Delta\tau_t = P_{PM}(t) - P_{PM}(t - \delta t) \quad (3)$$

A tradable event is triggered if $|\Delta\tau_t| > \gamma$.

3.5 Optimization Objective and Weights Allocation

Upon observing a valid signal, we construct a portfolio $\mathbf{w} = [w_1, \dots, w_N]^T$ to maximize exposure to the interest rate shock:

$$\max_{\mathbf{w}} \sum_{i=1}^N w_i \hat{\beta}_{i,\tau} \quad (4)$$

Subject to constraints:

$$\sum_{i=1}^N w_i = 0 \quad (\text{Dollar Neutrality}) \quad (5)$$

$$\sum_{i=1}^N w_i \hat{\beta}_{i,M} = 0 \quad (\text{Market Neutrality}) \quad (6)$$

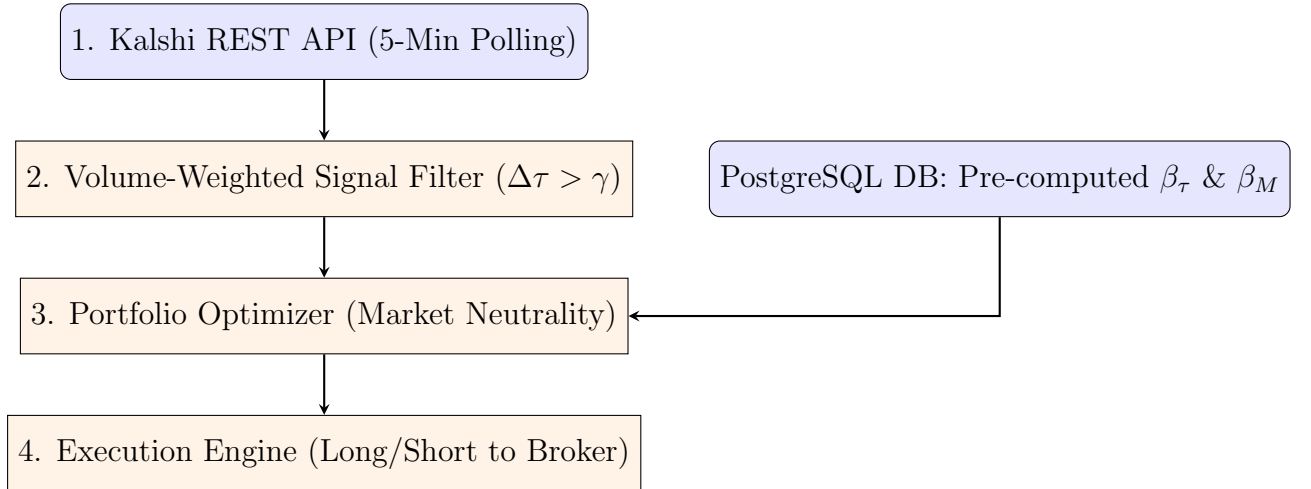
$$\sum_{i=1}^N |w_i| = 1 \quad (\text{Leverage Constraint}) \quad (7)$$

4 System Architecture and Data Pipeline

To implement this theoretical framework in a production environment, a robust data pipeline and modular code architecture are required. The system is designed to be cost-effective, leveraging containerized environments and polling mechanisms to comply with API rate limits.

4.1 Algorithm Data Pipeline

The decision-making process avoids the complexity of high-frequency WebSockets, utilizing a robust 5-minute polling mechanism. The workflow is illustrated below:



1. **Ingestion:** A scheduled job polls probability data from the Kalshi REST API every 5 minutes. This ensures compliance with free-tier API rate limits while maintaining sufficient resolution for medium-frequency trading.
2. **Filtering:** The system calculates the rolling probability delta. If the delta exceeds the threshold γ and is supported by real volume, a "Macro Shock Event" is triggered.
3. **Optimization:** The system queries a local PostgreSQL database for pre-calculated β_τ and β_M parameters. The optimizer mathematically solves for the target portfolio weights.

4. **Execution:** Orders are generated and sent to the broker API incorporating a programmed delay to bypass the initial ETF noise.

4.2 Code Repository Architecture

The system relies on a modular Python architecture, containerized via Docker for seamless deployment and portability. The repository distinctly separates offline research from the live execution engine:

```
project_root/
|-- data_ingestion/
|   |-- kalshi_poller.py          # REST API polling every 5 mins
|   |-- yfinance_updater.py      # Pulls daily/intraday equity data via yfinance
|-- research_and_backtesting/
|   |-- beta_calculator.py       # Offline OLS regressions for macro betas
|   |-- vectorbt_simulator.py    # Historical backtesting engine
|-- trading_engine/
|   |-- signal_generator.py       # Evaluates Kalshi Delta against noise
|   |-- portfolio_optimizer.py    # SciPy optimizer enforcing Neutrality
|   |-- execution_router.py      # Interfaces with the broker API
|-- database/
|   |-- init_postgres.sql        # PostgreSQL schema setup
|   |-- db_connector.py          # SQLAlchemy models and queries
|-- Dockerfile                   # Container instructions
|-- docker-compose.yml           # Orchestrates the Python App and Postgres DB
|-- main.py                      # The central orchestration script
```

4.3 Infrastructure and Deployment

To maintain a highly available yet cost-effective production environment, the entire architecture is containerized using Docker. The local stack utilizes `docker-compose` to spin up the Python execution environment alongside a PostgreSQL database, which stores historical β calculations, Kalshi probability logs, and order execution history.

For live deployment, the containerized application can be hosted on academic or student-tier cloud infrastructure. Platforms providing GitHub Student Developer Pack grants (such as DigitalOcean Droplets or Azure for Students) offer free computing credits. This allows the algorithm to run uninterrupted 24/7, managing the 5-minute Kalshi polling and execution routing without the need for expensive institutional hosting.

5 Target Universe and Empirical Testing Hierarchy

5.1 Macroeconomic Signal Prioritization: CPI, FED, and NFP

Supported by the empirical findings of Diercks et al. (2026), which demonstrate that Kalshi implied densities significantly outperform Bloomberg consensus in forecasting headline CPI [3], our testing methodology follows a specific hierarchy. This hierarchy is designed to capture the highest signal-to-noise ratio in macroeconomic shocks before they diffuse into individual equity prices.

The empirical evaluation is categorized by the following macro-betas:

1. **Inflation (CPI):** The primary focus, as Kalshi displays the strongest predictive edge here. The target universe is sorted by β_{CPI} (Inflation Beta).
2. **Interest Rates (FED):** Evaluated using β_r (Interest Rate Beta). This measures sensitivity to capital structure and discount rate shifts.
3. **Employment (NFP):** Evaluated using cyclical sensitivity measures.

5.2 Cross-Sectional Candidates (Example: Interest Rate Shock)

To illustrate the application of these ex-ante estimated betas—calculated offline during the nightly ingestion phase—we define a target universe representing the extremes of the sensitivity spectrum.

5.2.1 Short Candidates: High Sensitivity (e.g., RGTI, TSLA)

Companies with near-infinite duration, such as pre-revenue quantum computing firm Rigetti (RGTI), derive their valuation entirely from distant future cash flows, making them hyper-sensitive to discount rate increases [5]. Similarly, companies heavily reliant on consumer financing and leverage, such as Tesla (TSLA), face immediate demand destruction when rates rise. These form the core of our short portfolio during a rate-hike signal.

5.2.2 Long Candidates: Cash-Rich and NIM-Positive (e.g., GOOG, JPM)

Conversely, mega-cap technology firms with massive cash reserves and zero net debt, such as Alphabet (GOOG), effectively earn higher risk-free returns on their balance sheets when rates rise, offsetting operational headwinds. Traditional financial institutions like J.P. Morgan (JPM) benefit directly from Net Interest Margin (NIM) expansion. These serve as our long candidates, providing structural hedging and alpha potential against the short basket.

6 Backtesting Methodology and Infrastructure

6.1 Data Synchronization (Prediction Markets vs. Equities)

To prevent look-ahead bias and ensure accurate backtesting, data sources must be carefully synchronized. Equity market data is sourced seamlessly via the `yfinance` API, which provides robust daily pricing for historical beta calculations, as well as intra-day (e.g., 5-minute interval) pricing for execution simulation. This equity data is then timestamp-aligned with the 5-minute polling intervals from the Kalshi API. By standardizing both the signal generation and the equity pricing to 5-minute discrete blocks, we simulate realistic execution conditions and explicitly account for the REST API's inherent latency.

6.2 Signal Generation and Thresholding (γ)

A simple $\Delta\tau$ threshold is insufficient due to low liquidity in prediction markets. We introduce a Volume-Weighted Signal Filter: a probability shift is only considered valid if it is accompanied by a statistically significant spike in Order Flow Imbalance (OFI), confirming that the shift is driven by informed trading rather than isolated noise.

6.3 Execution Logic and Slippage Assumptions

To avoid competing with HFT index arbitrageurs, the algorithm imposes an artificial execution delay (e.g., $T_0 + 15$ minutes) following a signal. The holding period targets the cross-sectional convergence window, ranging from 2 to 72 hours. Conservative slippage and borrow-cost models are applied to the short leg of the portfolio.

7 Risk Management and Drawdown Scenarios

7.1 Correlation Breakdown in Black Swan Events

The primary risk to this strategy is a correlation breakdown. During severe macro-panics ("Black Swans"), the cross-sectional correlation of all equities tends toward 1.0. In such regimes, the fundamental β_τ differentiation breaks down, and the market neutrality constraint ($\sum w_i \beta_{i,M} = 0$) may fail to provide adequate hedging, necessitating global stop-loss mechanisms.

7.2 False Positives and Liquidity Shocks in PMs

Prediction markets are susceptible to manipulation by large singular trades ("whales"). A sudden multi-million dollar order can skew probabilities drastically, triggering a false macro signal. The volume and order-book depth filters outlined in Section 6.2 are critical to mitigating this risk.

8 Results and Discussion (Empirical Phase)

[This section is reserved for out-of-sample performance metrics, including Sharpe Ratio, Maximum Drawdown, and Alpha generation relative to the benchmark index, to be populated following the empirical backtesting phase.]

8.1 Out-of-Sample Performance Metrics

8.2 Conclusion and Future Work

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